

 Learn  Practice

Chemical bonding & hybridisation — Issue #33 candidate: Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Electron-domain geometry versus molecular shape

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Domain count before hybrid label

Question type: Constructed response

Source: Owner-supplied question

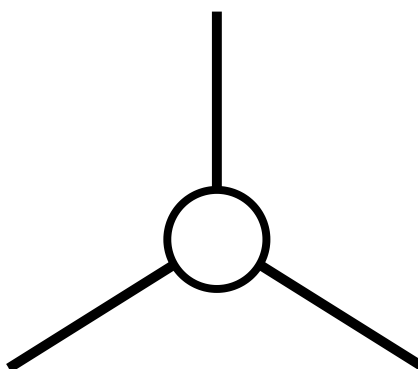
Provenance: Owner-supplied question

ATTEMPT FIRST

BF₃ and NH₃ each have three sigma bonds to the central atom. BF₃ has no central lone pair; NH₃ has one. Explain why three sigma bonds alone cannot determine the same central hybridisation and molecular shape for both.

Representation

Count directions / regions first



3 domain directions

Domain directions

Conditions

- Use the bounded introductory localized

Common wrong route · open if you want a warning

hybrid-orbital model
requested by the question;
do not treat molecular shape
alone as direct measurement
of orbital mixing.

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Common wrong route

Three sigma bonds always imply three domains.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Compare full local inventories, not only the common three sigma bonds.

H1 Representation

Guided

A central lone pair adds a domain even though it adds no bonded nucleus.

H2 First move

Guided

Make two columns: bonded neighbours and central lone pairs; connect totals to geometry yourself.

Need the concept again?

Electron-domain geometry
versus molecular shape

Answer and working

Hide

Reasoning route

1 Understand

Inventory bonds and central lone pairs separately.

Uses

BF₃ and NH₃ each have three sigma bonds to the central atom. BF₃ has no central lone pair; NH₃ has one. Explain why three sigma bonds alone cannot determine the same central hybridisation and molecular shape for both.

Why valid: The givens supply three bonds at each centre but different lone-pair counts.

Result: BF₃: 3+0=3 domains; NH₃: 3+1=4.

2 Connect

Key move

Apply the bounded introductory domain models.

Uses BF₃: 3+0=3 domains; NH₃: 3+1=4.

Why valid: The local total, rather than bonds alone, selects the model arrangement.

Result: BF₃: planar/sp²; NH₃: tetrahedral domains/sp³-like.

3 Calculate

Name shape using bonded nuclei only.

Uses BF₃: planar/sp²; NH₃: tetrahedral domains/sp³-like.

Why valid: The N lone-pair domain has no bonded nucleus.

Result: BF₃ is trigonal planar; NH₃ is trigonal pyramidal.

4 Interpret

Compare NH₃ with H₂O.

Uses BF₃ is trigonal planar; NH₃ is trigonal pyramidal.

Why valid: Three-bond/one-pair and two-bond/two-pair inventories both total four.

Result: The same domain total can give different bonded-nucleus shapes.

BF₃ has three B domains; NH₃ has three bonds plus one N lone pair. Introductory sp²/planar versus sp³-like domain/pyramidal shape follows the local model.

Independent check

Changed check: H₂O and NH₃ have different numbers of bonds but the same four-domain total.

Test the rule on a changed case

Count local domains and bonded nuclei separately in BF₃, NH₃ and H₂O.

Your prediction and reason

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

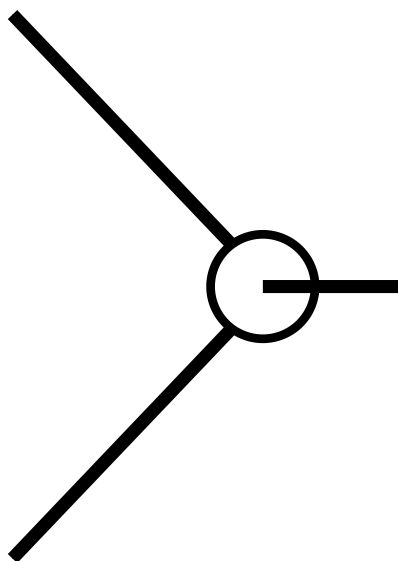
Provenance: Owner-supplied question

ATTEMPT FIRST

For H₂C=O, assign the carbon hybridisation, count the sigma and pi components of the C=O bond, and identify the carbon orbital left available for the pi component. Keep electron-domain counting separate from counting shared electron pairs.

Representation

Stage 1 — sigma-direction frame



page = molecular plane

Sigma-direction frame

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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Common wrong route

A carbon p function can be counted in a hybrid and as a residual orbital.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Conserve carbon orbitals while separating domains from sigma/pi components.

H1 Representation Guided

Three sigma directions do not consume the remaining perpendicular p orbital.

H2 First move Guided

Assign each carbon valence orbital once: hybrids to sigma, residual p to pi.

Need the concept again?

Sigma/pi orbital inventory
without double-counting

Answer and working

Hide

Reasoning route

1 Understand

Count C neighbours in HCHO.

Uses

For H2C=O, assign the carbon hybridisation, count the sigma and pi components of the C=O bond, and identify the carbon orbital left available for the pi component.
Keep electron-domain counting separate from counting shared electron pairs.

Why valid: Two H and one O are three connections; bond multiplicity is a separate inventory.

Result: Three domains; no C lone pair.

2 Connect

Key move

Allocate the four carbon valence functions.

Uses Three domains; no C lone pair.

Why valid: One 2s plus two 2p form three sp² hybrids, leaving the third 2p unused in that mixing.

Result: Three sigma-direction hybrids plus one residual p.

3 Calculate

Resolve the C=O bond components.

Uses Three sigma-direction hybrids plus one residual p.

Why valid: A localized double-bond account has one axial sigma and one lateral pi component.

Result: C=O: one sigma + one pi; residual C p supplies the pi role.

4 Interpret

Close the basis ledger using an sp comparison.

Uses C=O: one sigma + one pi; residual C p supplies the pi role.

Why valid: sp gives two hybrids and two residual p functions from the same four inputs.

Result: Both allocations total four; no p function is spent twice.

HCHO carbon has three-domain sp² accounting, three sp² sigma directions and one unhybridised p orbital; C=O has one sigma plus one pi.

Independent check

Changed check: for an sp carbon, two hybrids plus two residual p orbitals still total four basis functions.

Test the rule on a changed case

Inventory sp² carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p?

Your prediction and reason

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

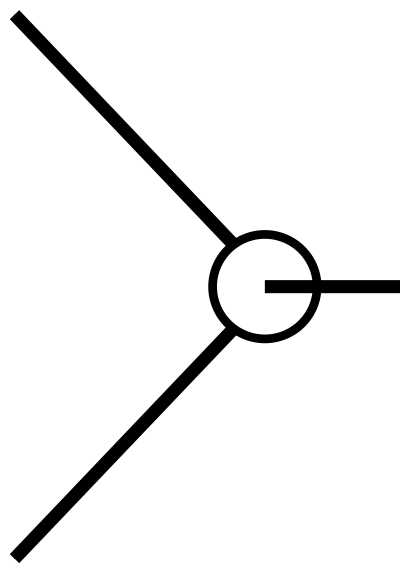
Provenance: Owner-supplied question

ATTEMPT FIRST

Ethene is $\text{H}_2\text{C}=\text{CH}_2$. Translate this Lewis representation into a qualitative orbital-overlap diagram showing the C–C sigma framework, the two unhybridised carbon p orbitals and their side-on overlap. Label which directions are in the molecular plane and which are perpendicular to it.

Representation

Stage 1 — sigma-direction frame



page = molecular plane

Sigma-direction frame

Conditions

- Use the bounded introductory localized

Common wrong route · open if you want a warning

hybrid-orbital model
requested by the question;
do not treat molecular shape
alone as direct measurement
of orbital mixing.

Common wrong route

Page orientation is the same as relative
orbital orientation.

Need a hint? · guided support

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Construct the relation between the
molecular plane and the pi orbitals.

H1 Representation

Guided

Three sigma directions lie in each sp^2
carbon plane; residual p is
perpendicular to that plane.

H2 First move

Guided

Draw both carbon frames, then align
the residual p directions for possible
lateral overlap.

Need the concept again?

Sigma/pi orbital inventory
without double-counting

Answer and working

Hide

Reasoning route

1 Understand

Build the two local carbon sigma frames.

Uses

Ethene is $\text{H}_2\text{C}=\text{CH}_2$. Translate this Lewis representation into a qualitative orbital-overlap diagram showing the C–C sigma framework, the two unhybridised carbon p orbitals and their side-on overlap. Label which directions are in the molecular plane and which are perpendicular to it.

Why valid: Each ethene carbon has three sigma directions and one remaining p function in sp^2 accounting.

Result: Two in-plane three-direction frames; one perpendicular p at each C.

2 Connect

Key move

Align the p functions relative to the molecular plane.

Uses Two in-plane three-direction frames; one perpendicular p at each C.

Why valid: A residual p is perpendicular to its carbon sigma plane.

Result: Parallel perpendicular p directions in planar ethene.

3 Calculate

Draw both axial and lateral overlap.

Uses Parallel perpendicular p directions in planar ethene.

Why valid: The C–C sigma component is axial; lateral p overlap lies above and below the plane.

Result: A sigma framework plus a pi component with a node in the molecular plane.

4 Interpret

Compare rigid rotation with relative CH_2 -group rotation.

Uses A sigma framework plus a pi component with a node in the molecular plane.

Why valid: Only the latter changes relative p alignment.

Result: Whole-picture rotation preserves overlap geometry; relative twisting can reduce pi alignment.

Ethene has a coplanar sigma framework; the two residual p directions are parallel and perpendicular to it in the introductory model.

Independent check

Changed check: rotate one CH₂ group while preserving C-C connectivity; ask which overlap is sensitive to the relative orientation.

Test the rule on a changed case

Rotate the whole ethene picture, then rotate only one CH₂ group. Which operation changes relative p alignment?

Your prediction and reason

COMPETITION TRANSFER

Close an orbital inventory under a changed local description

Original transfer task; not claimed as an official exam/PYQ item.

A neutral carbon centre is described by three coplanar sigma-bond directions and one perpendicular p-orbital participation. Without using a memorised molecule name, give an orbital-job inventory and state one check that would expose double-counting.

Check your reasoning

Three sigma-framework hybrid directions and one separate p-orbital pi job; check that each local valence orbital is allocated once.

1. The three sigma directions define the local sigma framework.
2. The perpendicular p direction is reserved for side-on pi overlap.

3. A closure check prevents any orbital from being assigned twice.

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CO₂, connect the central-carbon Lewis domain count to its hybrid-orbital model, the number of remaining carbon p orbitals and the sigma/pi composition of its two C=O bonds. Explain how these pieces fit one consistent linear arrangement.

Representation

Stage 1 — two opposite domain /



Linear sigma axis

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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Common wrong route

A localized CO₂ drawing uniquely measures the terminal oxygen basis.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

The central carbon inventory must support two directions and two pi components.

H1 Representation

Guided

sp accounting leaves two mutually perpendicular p directions.

H2 First move

Guided

Allocate carbon basis functions and bond components in separate ledgers; keep the localized drawing explicitly a model.

Need the concept again?

Sigma/pi orbital inventory
without double-counting

Answer and working

Hide

Reasoning route

1

Understand

Count two O neighbours at CO₂ carbon.

Uses

For CO₂, connect the central-carbon Lewis domain count to its hybrid-orbital model, the number of remaining carbon p orbitals and the sigma/pi composition of its two C=O bonds. Explain how these pieces fit one consistent linear arrangement.

Why valid: Each C=O multiple bond contributes one local direction.

Result: Two carbon domains; linear introductory framework.

2 Connect

Key move

Allocate carbon orbitals without duplication.

Uses Two carbon domains; linear introductory framework.

Why valid: One 2s plus one 2p yields two sp hybrids; two p directions remain perpendicular.

Result: Two sigma directions and two residual C p functions.

3 Calculate

Account for each localized C=O component.

Uses Two sigma directions and two residual C p functions.

Why valid: Two double bonds require two sigma and two pi components in this bookkeeping model.

Result: Two sigma + two pi; all four carbon functions have declared roles.

4 Interpret

Bound the inference about terminal oxygen.

Uses Two sigma + two pi; all four carbon functions have declared roles.

Why valid: A localized basis drawing is one representation of the electronic description; carbon counting does not uniquely measure an oxygen basis.

Result: Linear carbon accounting is consistent; a unique observed O orbital-plane picture is not established.

CO₂ carbon: two domains/sp, two sigma components and two pi components. A localized VB drawing is useful accounting, not a unique experimentally compulsory terminal-O orbital-plane picture.

Independent check

Changed check: HCN carbon also has two sigma directions and two residual p directions although its neighbours differ.

Test the rule on a changed case

Give carbon orbital inventory for CO₂ and HCN. Does a localized drawing uniquely measure the oxygen orbital basis?

Your prediction and reason

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Domain count before hybridisation label

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Domain count before hybrid label

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

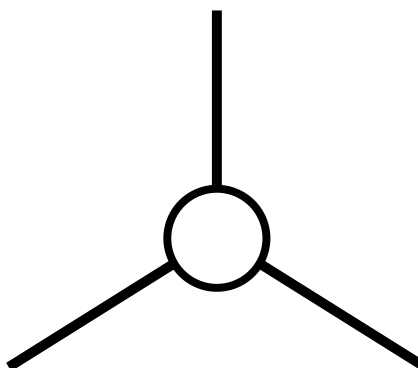
ATTEMPT FIRST

Consider the claim: 'Any central atom with three sigma bonds must be sp².' Given BF₃: three B–F sigma bonds, zero boron lone pairs, planar; NH₃: three N–H sigma bonds, one nitrogen lone pair, pyramidal. Decide whether these observations support the claim, give the decisive

Representation

warrant, and state a more defensible introductory rule. Explain what the shape observations alone do not prove about orbitals.

Count directions / regions first



3 domain directions

Domain directions

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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Common wrong route

Three sigma bonds suffice to prove sp^2 .

Need a hint? · guided support

—

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Test whether three sigma bonds are sufficient for the proposed label.

H1 Representation

Guided

A counterexample must hold the claimed condition while changing an omitted relevant condition.

H2 First move

Guided

Choose a centre with three bonds and a central lone pair; test the claim rather than naming a familiar molecule only.

Need the concept again?

Domain count before hybridisation label

Answer and working

Hide

Reasoning route

1 Understand

Hold the claimed three-sigma condition fixed.

Uses

Consider the claim: 'Any central atom with three sigma bonds must be sp^2 .' Given BF₃: three B–F sigma bonds, zero boron lone pairs, planar; NH₃: three N–H sigma bonds, one nitrogen lone pair, pyramidal. Decide whether these observations support the claim, give the decisive warrant, and state a more defensible introductory rule. Explain what the shape observations alone do not prove about orbitals.

Why valid: Both supplied centres satisfy it, so an omitted lone-pair condition can test the implication.

Result: BF₃ and NH₃ both have three sigma bonds.

2 Connect

Key move

Use nitrogen as the counterexample.

Uses BF₃ and NH₃ both have three sigma bonds.

Why valid: NH₃ also has one central lone pair; its local total is four.

Result: Three sigma bonds alone do not imply three domains/sp².

3

Calculate

Replace the claim with a conditional local rule.

Uses Three sigma bonds alone do not imply three domains/sp².

Why valid: Within localized introductory classification, bonds plus central lone pairs determine the domain total.

Result: Use the full inventory: three domains -> sp² model; four -> sp³-like model.

4

Interpret

Separate shape observations from a unique orbital basis.

Uses Use the full inventory: three domains -> sp² model; four -> sp³-like model.

Why valid: Nuclear geometry constrains a description but does not measure hybrid coefficients.

Result: The counterexample rejects the bond-only rule without proving unique physical hybrid orbitals.

NH₃ falsifies three-sigma-implies-sp²: its local lone pair makes four domains. Geometry constrains a model but does not uniquely prove a hybrid basis.

Independent check

Changed check: BF₃ and NH₃ share three bonds but differ in local lone pairs; the proposed rule predicts the same result incorrectly.

Test the rule on a changed case

Would a three-bond nitrogen with a lone pair and a three-bond boron without one require identical domain inventories? Explain.

Your prediction and reason

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Domain count before hybridisation label

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Domain count before hybrid label

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In CH₃F, carbon has three C–H single bonds and one C–F single bond. Assign its introductory local hybridisation and explain why the number of electron-domain directions is unchanged from CH₄ even though the attached atoms are different. Do not assume all bond properties are identical.

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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Common wrong route

An electronegative substituent adds a domain beyond its bond direction.

Need a hint? · guided support

—

**HINT LADDER · REVEAL
ONLY WHAT YOU NEED**

Guided support

H0 Key concept

Guided

Separate number of bonding directions from bond identity or strength.

H1 Representation

Guided

Replacing a substituent changes chemical environment without automatically changing connectivity count.

H2 First move

Guided

Recount neighbours before considering any unprovided property differences.

Need the concept again?

Domain count before hybridisation label

Answer and working

Hide

Reasoning route

1

Understand

Recount carbon neighbours after H-to-F substitution.

Uses

In CH₃F, carbon has three C–H single bonds and one C–F single bond. Assign its introductory local hybridisation and explain why the number of electron-domain directions is unchanged from CH₄ even though the attached atoms are different. Do not assume all bond properties are identical.

Why valid: CH₃F has three H and one F directly attached to C.

Result: Four carbon bonding directions.

2

Connect

Key move

Apply the same local four-domain allocation.

Uses Four carbon bonding directions.

Why valid: Changing substituent identity does not itself create a new neighbour or lone pair at C.

Result: Introductory sp³ carbon model.

3

Calculate

State what the count leaves open.

Uses Introductory sp³ carbon model.

Why valid: Bond lengths, polarity and exact angles require information beyond neighbour counting.

Result: No equality of C–H and C–F properties is inferred.

4

Interpret

Test CH₂F₂ as a changed inventory.

Uses No equality of C–H and C–F properties is inferred.

Why valid: Two H and two F are still four neighbours.

Result: Four domains remain; electronegativity is not an extra domain.

CH₃F carbon has four bonding directions and introductory sp³ accounting. The count alone establishes neither bond lengths nor their detailed causes.

Independent check

Changed check: CH₂F₂ still has four central bonding directions; electronegativity is not an extra domain.

Test the rule on a changed case

Count carbon domains in CH_4 , CH_3F and CH_2F_2 . Does each F add an extra domain beyond its C-F direction?

Your prediction and reason

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

HCN is represented as $\text{H}-\text{C}\equiv\text{N}$ with a lone pair on nitrogen. Assign the introductory hybridisation at carbon and nitrogen, and count the sigma and pi components of the $\text{C}\equiv\text{N}$ bond. Explain how the local domain count is compatible with those components.

Representation

Stage 1 — two opposite domain /



Linear sigma axis

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

—

Common wrong route

Three C-N bond lines supply three C-centred directions.

Need a hint? · guided support

—

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Keep local inventories separate from the triple-bond component inventory.

H1 Representation

Guided

One bonded neighbour supplies one direction even when three bond components are present.

H2 First move

Guided

At C and N count local sigma directions/lone pairs, then tally one sigma plus residual pi components.

Need the concept again?

Sigma/pi orbital inventory
without double-counting

Reasoning route

1 Understand

Inventory carbon and nitrogen separately in HCN.

Uses

HCN is represented as $\text{H}-\text{C}\equiv\text{N}$ with a lone pair on nitrogen. Assign the introductory hybridisation at carbon and nitrogen, and count the sigma and pi components of the $\text{C}\equiv\text{N}$ bond. Explain how the local domain count is compatible with those components.

Why valid: C has H and N neighbours; N has one C neighbour plus one lone pair.

Result: C: two domains; N: two domains.

2 Connect

Key move

Allocate each local four-function valence basis.

Uses C: two domains; N: two domains.

Why valid: Two sp hybrids leave two perpendicular p functions at each second-period centre.

Result: Introductory sp at C and N.

3 Calculate

Count triple-bond components without adding local directions.

Uses Introductory sp at C and N.

Why valid: Axial overlap gives one sigma; the two residual p pairs give two pi components.

Result: C-N: one sigma + two pi; C-H adds one sigma.

4 Interpret

Close the ten-valence-electron inventory.

Uses C-N: one sigma + two pi; C-H adds one sigma.

Why valid: $\text{H}+\text{C}+\text{N}$ contributes $1+4+5=10$; four shared pairs and one N lone pair use five pairs.

Result: Eight bonding electrons + two lone-pair electrons = ten.

HCN carbon and nitrogen have introductory sp local accounting; C-N contains one sigma and two pi components.

Independent check

Changed check: total valence electrons HCN = $1+4+5 = 10$; the three shared pairs and N lone pair plus C-H pair use all ten.

Test the rule on a changed case

Compare domain counts at C and N with the total number of shared pairs in HCN.

Your prediction and reason

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Electron-domain geometry versus molecular shape

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Electron-domain geometry versus molecular shape

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

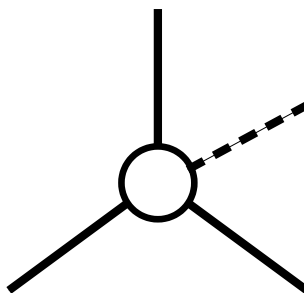
For methylamine, $\text{CH}_3\text{-NH}_2$, carbon has four single bonds and nitrogen has three single bonds plus a lone pair.

Compare the introductory local hybridisation and arrangement

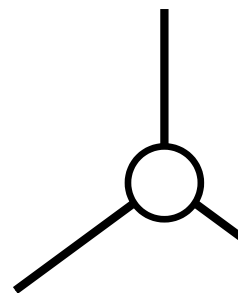
Representation

of bonded atoms at carbon and nitrogen. Explain why the same hybridisation label does not require identical molecular shapes.

Same four-domain scaffold



4 bond positions



3 bonds + 1 lone pair

Occupancy cases

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route • open if you want a warning

—

Common wrong route

One whole-molecule label determines both carbon and nitrogen shapes.

Need a hint? • guided support

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HINT LADDER • REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Count local domains and bonded nuclei at two different centres.

H1 Representation

Guided

A lone pair contributes to electron-domain geometry but is excluded from molecular-shape naming.

H2 First move

Guided

Use separate C and N inventories; do not carry a whole-molecule label from one atom to the other.

Need the concept again?

Electron-domain geometry
versus molecular shape

Answer and working

Hide

Reasoning route**1 Understand**

Make atom-local inventories in CH_3NH_2 .

Uses

For methylamine, $\text{CH}_3\text{-NH}_2$, carbon has four single bonds and nitrogen has three single bonds plus a lone pair. Compare the introductory local hybridisation and arrangement of bonded atoms at carbon and nitrogen. Explain why the same hybridisation label does not require identical molecular shapes.

Why valid: C has four single bonds; N has three single bonds and one lone pair.

Result: C: $4+0$; N: $3+1$; both total four domains.

2 Connect

Key move

Assign the localized introductory domain models.

Uses C: $4+0$; N: $3+1$; both total four domains.

Why valid: Each four-domain centre can use sp^3 -like accounting under the stated local model.

Result: Four model directions at each centre.

3 Calculate

Count nuclei for the separate shape descriptions.

Uses Four model directions at each centre.

Why valid: C has four bonded neighbours while N has three; a lone pair is not an atomic vertex.

Result: Tetrahedral C nuclear arrangement; pyramidal N arrangement.

4 Interpret

Replace an N-H bond with an N-CH₃ bond in a neutral amine comparison.

Uses Tetrahedral C nuclear arrangement; pyramidal N arrangement.

Why valid: The N inventory stays three bonds plus one lone pair.

Result: Local N count and pyramidal introductory description persist; exact angle is unresolved.

Methylamine C has four bonded directions; N has three bonded directions and one lone pair. Introductory tetrahedral C and pyramidal N nuclear arrangements differ despite four-domain totals.

Independent check

Changed check: replacing N-H with N-CH₃ preserves the local three-bond-plus-lone-pair N inventory in the neutral localized amine model.

Test the rule on a changed case

Compare local N inventory with carbon inventory in CH₃NH₂. Which centre includes a lone-pair direction?

Your prediction and reason

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

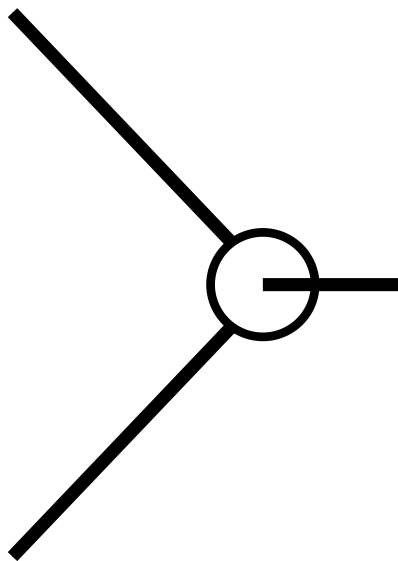
Provenance: Owner-supplied question

ATTEMPT FIRST

A carbon centre has three sigma-bond directions, no lone pair, and participates in one pi bond. Translate this description into a qualitative local orbital arrangement and check that the orbitals used in the sigma and pi parts have not been counted twice.

Representation

Stage 1 — sigma-direction frame



page = molecular plane

Sigma-direction frame

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape

Common wrong route · open if you want a warning

—

alone as direct measurement of orbital mixing.

Common wrong route

An input p function can be spent twice in the orbital inventory.

Need a hint? · guided support

—

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Avoid spending a carbon basis function twice.

H1 Representation

Guided

For second-period carbon, one 2s and three 2p functions form a four-dimensional valence basis.

H2 First move

Guided

Assign three hybrid sigma directions and reserve only unused functions for the pi job.

Need the concept again?

Sigma/pi orbital inventory without double-counting

Answer and working

Hide

Reasoning route

1 Understand

Start from four carbon valence-basis functions.

Uses

A carbon centre has three sigma-bond directions, no lone pair, and participates in one pi bond. Translate this description into a qualitative local orbital arrangement and check that the orbitals used in the sigma and pi parts have not been counted twice.

Why valid: The second-period basis is one 2s plus three 2p.

Result: Four inputs; three sigma directions and one pi role required.

2 Connect

Key move

Mix only the functions assigned to sigma directions.

Uses Four inputs; three sigma directions and one pi role required.

Why valid: One s plus two p makes three in-plane sp² hybrids.

Result: Three hybrids; one p remains outside the mixing.

3 Calculate

Assign the residual p to the pi role.

Uses Three hybrids; one p remains outside the mixing.

Why valid: It is perpendicular to the sigma plane and can overlap laterally with a suitable neighbour.

Result: Three sigma hybrids + one residual perpendicular p.

4 Interpret

Check against sp and sp³ allocations.

Uses Three sigma hybrids + one residual perpendicular p.

Why valid: Those alternatives account for two hybrids plus two p, or four hybrids plus zero p.

Result: All totals are four; no function is simultaneously a hybrid input and an extra residual p.

sp² accounting is three hybrid orbitals plus one residual p. A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

Independent check

Changed check: sp accounting gives two hybrids plus two residual p; sp³ gives four hybrids plus zero residual p.

Test the rule on a changed case

Build an allocation ledger for sp, sp², sp³ and flag any duplicate-use entry.

Your prediction and reason

COMPETITION TRANSFER

Close an orbital inventory under a changed local description

Original transfer task; not claimed as an official exam/PYQ item.

A neutral carbon centre is described by three coplanar sigma-bond directions and one perpendicular p-orbital participation. Without using a memorised molecule name, give an orbital-job inventory and state one check that would expose double-counting.

Check your reasoning

Three sigma-framework hybrid directions and one separate p-orbital pi job; check that each local valence orbital is allocated once.

1. The three sigma directions define the local sigma framework.
2. The perpendicular p direction is reserved for side-on pi overlap.
3. A closure check prevents any orbital from being assigned twice.

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Electron-domain geometry versus molecular shape

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Electron-domain geometry versus molecular shape

Question type: Constructed response

Source: Owner-supplied question

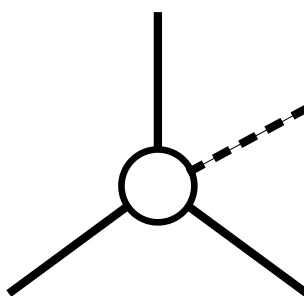
Provenance: Owner-supplied question

ATTEMPT FIRST

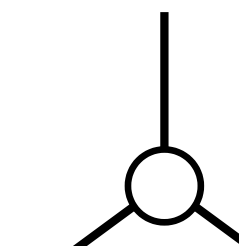
A student says, 'If the electron-domain geometry is tetrahedral, the atoms must form a tetrahedral molecular shape.' Evaluate the statement using CH_4 , NH_3 and H_2O , with their respective central lone-pair counts of zero, one and two.

Representation

Same four-domain scaffold



4 bond positions



3 bonds + 1 lone pair

Occupancy cases

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route • open if you want a warning

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Common wrong route

Four-domain geometry equals the atom-only shape in every case.

Need a hint? • guided support

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HINT LADDER • REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept Guided

Test whether domain geometry and nuclear shape are interchangeable.

H1 Representation Guided

A lone-pair domain has no bonded nucleus to include in molecular-shape naming.

H2 First move Guided

Hold four domains fixed and vary how many slots are bonded atoms.

Need the concept again?

Electron-domain geometry
versus molecular shape

Answer and working

Hide

Reasoning route

1 Understand

Hold the four-domain total fixed across three cases.

Uses

A student says, 'If the electron-domain geometry is tetrahedral, the atoms must form a tetrahedral molecular shape.' Evaluate the statement using CH₄, NH₃ and H₂O, with their respective central lone-pair counts of zero, one and two.

Why valid: CH₄, NH₃ and H₂O have respective bond/pair inventories 4/0, 3/1 and 2/2.

Result: All have four introductory domains.

2 Connect

Key move

Use all domains for domain geometry.

Uses All have four introductory domains.

Why valid: Bonding and lone-pair entries occupy the domain-model directions.

Result: Tetrahedral electron-domain arrangement in all three.

3 Calculate

Use bonded nuclei for molecular shape.

Uses Tetrahedral electron-domain arrangement in all three.

Why valid: Removing LP entries from the shape name changes the number of atomic vertices.

Result: CH₄ tetrahedral; NH₃ pyramidal; H₂O bent.

4 Interpret

State the condition for the two names to coincide.

Uses CH₄ tetrahedral; NH₃ pyramidal; H₂O bent.

Why valid: When there is no central lone pair every domain direction terminates at a bonded nucleus.

Result: The student claim holds for the no-lone-pair case, not universally.

CH₄, NH₃, H₂O have four introductory domains but tetrahedral, pyramidal and bent nuclear shapes respectively.

Independent check

Changed check: a four-bond/no-lone-pair centre cannot be assigned the same atom-only shape as a two-bond/two-lone-pair centre.

Test the rule on a changed case

For four-domain inventories with 4,3,2 bonded atoms, give the atom-only shape and explain omitted slots.

Your prediction and reason

COMPETITION TRANSFER

Hold domain geometry fixed while changing atom occupancy

Original transfer task; not claimed as an official exam/PYQ item.

A four-domain centre is changed from four bonds to two bonds plus two lone pairs without changing the total domain count. Explain which geometry description is preserved and which atom-only description must be reconsidered.

Check your reasoning

The four-domain electron geometry stays tetrahedral; the molecular shape must be derived from the two bonded-atom positions.

1. All four electron domains remain.
2. Molecular shape omits lone-pair positions.